

CHAPTER – XII

APPENDIX

12.1 – Source Code

app.py

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/dashboard01')
def dashboard01():
    return render_template('dashboard01.html')

@app.route('/dashboard02')
def dashboard02():
    return render_template('dashboard02.html')

@app.route('/story')
def story():
    return render_template('story.html')

if __name__ == '__main__':
    app.run(debug=True)
```

Base.py

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>{% block title %}Heart Disease Analysis{% endblock
%}</title>

    <!-- Bootstrap CSS -->
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootst
rap.min.css"
```

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```
rel="stylesheet">

<!-- Font Awesome -->
<link rel="stylesheet"
      href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.5.2/css/all.min.css">

<!-- Custom CSS -->
<link rel="stylesheet"
      href="{{ url_for('static', filename='css/style.css')
}}">

<style>

    body {
        font-family: 'Segoe UI', sans-serif;
        background-color: #f8f9fa;
        padding-top: 80px;
    }

    /* Fixed Navbar */
    .navbar-custom {
        background: linear-gradient(90deg, #dc3545, #ff6b6b);
        box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
    }

    .navbar-brand {
        font-size: 1.5rem;
        font-weight: 700;
        color: white !important;
    }

    .navbar-nav .nav-link {
        color: white !important;
        font-weight: 500;
        margin-left: 15px;
        transition: all 0.3s ease;
        border-radius: 8px;
        padding: 8px 15px !important;
    }

    .navbar-nav .nav-link:hover {
        background-color: rgba(255, 255, 255, 0.2);
        color: #fff !important;
    }

    .navbar-nav .nav-link.active {
        background-color: white;
        color: #dc3545 !important;
    }
```

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```
}

/* Hero Section */
.hero-section {
  background: linear-gradient(rgba(220,53,69,0.8),
    rgba(220,53,69,0.8)),
    url("https://images.unsplash.com/photo-
1576091160399-112ba8d25d1a?w=1600");
  background-size: cover;
  background-position: center;
  min-height: 90vh;
  display: flex;
  align-items: center;
}

/* Feature Cards */
.feature-card {
  border: none;
  border-radius: 20px;
  transition: 0.4s;
}

.feature-card:hover {
  transform: translateY(-10px);
}

/* Dashboard Cards */
.dashboard-card {
  border-radius: 20px;
  overflow: hidden;
}

/* Footer */
footer {
  background: #212529;
  color: white;
  padding: 20px 0;
  margin-top: 50px;
}

</style>

</head>

<body>

  <!-- Fixed Horizontal Navbar -->
  <nav class="navbar navbar-expand-lg navbar-dark navbar-custom
fixed-top">
```

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```
<div class="container">

  <a class="navbar-brand"
    href="{{ url_for('home') }}">

    <i class="fas fa-heartbeat"></i>
    Heart Disease Analysis

  </a>

  <button class="navbar-toggler"
    type="button"
    data-bs-toggle="collapse"
    data-bs-target="#navbarNav">

    <span class="navbar-toggler-icon"></span>

  </button>

  <div class="collapse navbar-collapse"
    id="navbarNav">

    <ul class="navbar-nav ms-auto">

      <li class="nav-item">
        <a class="nav-link"
          href="{{ url_for('home') }}">

          <i class="fas fa-home"></i>
          Home

        </a>
      </li>

      <li class="nav-item">
        <a class="nav-link"
          href="{{ url_for('about') }}">

          <i class="fas fa-info-circle"></i>
          About

        </a>
      </li>

      <li class="nav-item">
        <a class="nav-link"
          href="{{ url_for('dashboard01') }}">
```

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```
<i class="fas fa-chart-line"></i>
Dashboard 1

</a>
</li>

<li class="nav-item">
  <a class="nav-link"
    href="{{ url_for('dashboard02') }}">

    <i class="fas fa-chart-pie"></i>
    Dashboard 2

  </a>
</li>

<li class="nav-item">
  <a class="nav-link"
    href="{{ url_for('story') }}">

    <i class="fas fa-book-open"></i>
    Story

  </a>
</li>
</ul>
</div>
</div>

</nav>

<!-- Main Content -->
<main>

  {% block content %}
  {% endblock %}

</main>

<!-- Footer -->
<footer>

  <div class="container text-center">

    <h5>
      <i class="fas fa-heartbeat text-danger"></i>
```

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```
Heart Disease Analysis
</h5>

<p class="mb-0">
    An Interactive Healthcare Analytics Dashboard
using Flask,
    Bootstrap, and Data Visualization.
</p>

<small>
    © 2026 Heart Disease Analysis Project. All Rights
Reserved.
</small>

</div>

</footer>

<!-- Bootstrap JS -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstra
p.bundle.min.js">
</script>

</body>
</html>
```

Index.html

```
{% extends "base.html" %}

{% block title %}Home - Heart Disease Analysis{% endblock %}

{% block content %}

<section class="welcome-section py-5">
    <div class="container">

        <div class="row align-items-center">

            <!-- Left Side Content -->
            <div class="col-lg-6 mb-4 mb-lg-0">
                <h2 class="welcome-title mb-4">
                    Welcome
                </h2>
```

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```
<p class="welcome-text">
Heart disease (heart disease) is a group of
diseases related
to cardiovascular diseases, manifested by a
violation of the
normal functioning of the heart. May be caused
by damage to
the epicardium, pericardium, myocardium,
endocardium, valvular
apparatus of the heart, heart vessels.
</p>
```

```
<p class="welcome-text">
According to the National Heart, Lung and
Blood Institute in
Framingham (USA), the most important factors
in the development
of cardiovascular disease in humans are
obesity, sedentary
lifestyle and smoking.
</p>
```

```
</div>
<!-- Right Side Image -->
<div class="col-lg-6 text-center">

</div>
```

```
</div>
</div>
</section>
```

```
<div class="container my-5">
```

```
<div class="row g-4">
```

```
<div class="col-md-4">
```

```
<div class="card feature-card card-1 text-center p-4">
```

```
<div class="card-body">
```

```
<i class="fas fa-heartbeat"></i>
```

Heart Disease Analysis using Tableau

```
<h3 class="mt-3">Risk Factors</h3>

<p>
    Analyze smoking, obesity,
    diabetes, and lifestyle impacts.
</p>

</div>

</div>

</div>

<div class="col-md-4">

    <div class="card feature-card card-2 text-center p-4">

        <div class="card-body">

            <i class="fas fa-chart-line"></i>

            <h3 class="mt-3">Interactive Analysis</h3>

            <p>
                Explore dashboards for
                meaningful healthcare insights.
            </p>

        </div>

    </div>

</div>

</div>

<div class="col-md-4">

    <div class="card feature-card card-3 text-center p-4">

        <div class="card-body">

            <i class="fas fa-user-md"></i>

            <h3 class="mt-3">Better Decisions</h3>

            <p>
                Support evidence-based healthcare
                planning and interventions.
            </p>

        </div>

    </div>

</div>
```



```
        </div>

    </div>

</div>

</div>

</div>

{% endblock %}
```

about.html

```
{% extends "base.html" %}

{% block title %}About{% endblock %}

{% block content %}

<div class="container py-5">

    <div class="mt-5">

        <h1 class="text-danger mb-4">
            About the Project
        </h1><!-- Project Overview Section -->
<section class="py-5" style="background-color: #f8fbff;">
    <div class="container">

        <!-- Section Heading -->
        <div class="text-center mb-5">
            <h2 class="fw-bold text-primary">Project Overview</h2>
            <p class="lead text-muted">
                The <strong>Heart Disease Analysis</strong>
project aims to analyze patient health data
                to identify risk factors associated with heart
disease. Through interactive dashboards
                and visual analytics, the project helps healthcare
professionals understand trends,
```

Heart Disease Analysis using Tableau

support decision-making, and promote early detection of heart-related conditions.

```
</p>
</div>

<!-- Team Details Heading -->
<div class="text-center mb-4">
  <h3 class="fw-bold text-primary">Our Team</h3>
  <p class="text-muted">Meet the members behind this
project.</p>
</div>

<!-- Team Cards -->
<div class="row g-4">

  <!-- Team Member 2-->
  <div class="col-md-4">
    <div class="card shadow-sm border-0 h-100 text-
center">
      <div class="card-body">

        <h5 class="card-title fw-
bold">Thondamanadu Bharathi</h5>
        <p class="text-primary fw-semibold">Team
leader</p>

        <p class="card-text text-muted">
          Responsible for project planning,
coordination,
          and overall implementation of the
Heart Disease Analysis system.
        </p>

        <p class="mb-1"><strong>Email:</strong>
bharathi256009@zohomail.in</p>
        <p><strong>Roll No:</strong> 256009</p>
      </div>
    </div>
  </div>

  <!-- Team Member 2 -->
  <div class="col-md-4">
    <div class="card shadow-sm border-0 h-100 text-
center">
      <div class="card-body">

        <h5 class="card-title fw-bold">Dechepalli
```

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```

</div>
<div class="col-md-4">
  <div class="card shadow-sm border-0 h-100 text-center">
    <div class="card-body">
      <h5 class="card-title fw-bold">Nimmanapalli kavya</h5>
      <p class="text-primary fw-semibold">Team Member</p>
      <p class="card-text text-muted">
        Developed the Flask web application
        using
        HTML, CSS, Bootstrap, and integrated
        project components.
      </p>
      <p class="mb-1"><strong>Email:</strong>
        kavya256004@zohomail.in</p>
      <p><strong>Roll No:</strong> 256004</p>
    </div>
  </div>
</div>
</div>
</section>
</div>
</div>

```

Heart Disease Analysis using Tableau

```
{% endblock %}
```

Dashboard01.html

```
{% extends 'base.html' %}

{% block title %}Dashboard 01{% endblock %}

{% block content %}

<div class="container-fluid py-5">

    <div class="text-center mb-4">
        <h1 class="fw-bold text-danger">
            HEART DISEASE ANALYSIS Dashboard 01
        </h1>
        <p class="text-muted">
            Interactive visualization of Heart Disease Analysis
        </p>
    </div>

    <div class="card shadow-lg border-0 rounded-4">
        <div class="card-body p-3">

            <div class="tableau-container">
                <div class='tableauPlaceholder'
id='viz1781346546370' style='width:100%; height:900px;'>

                    <noscript>
                        <a href='#'>
                            <img alt='Dashboard01'

src='https://public.tableau.com/static/images/HE/HEARTDISEASEANALY
SISUSINGTABLAUE-DASHBOARD01/Dashboard01/1_rss.png'
                            style='border: none; width:100%;'>
                        </a>
                    </noscript>

                    <object class='tableauViz'


```

Heart Disease Analysis using Tableau

```
style='display:none; '>

        <param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F' />

        <param name='embed_code_version' value='3'

/>

        <param name='site_root' value='' />

        <param name='name'

value='HEARTDISEASEANALYSISUSINGTABLAUE-DASHBOARD01/Dashboard01'
/>

        <param name='tabs' value='no' />

        <param name='toolbar' value='yes' />

        <param name='animate_transition'

value='yes' />

        <param name='display_static_image'

value='yes' />

        <param name='display_spinner' value='yes'

/>

        <param name='display_overlay' value='yes'

/>

        <param name='display_count' value='yes' />

        <param name='language' value='en-US' />

        <param name='filter' value='publish=yes'

/>

        </object>
    </div>
</div>

</div>

<script type='text/javascript'>
    var divElement = document.getElementById('viz1781346546370');
```

Heart Disease Analysis using Tableau

```
var vizElement = divElement.getElementsByTagName('object')[0];

// Make dashboard responsive
vizElement.style.width = '100%';
vizElement.style.height = '900px';

if (window.innerWidth < 768) {
    vizElement.style.height = '700px';
}

var scriptElement = document.createElement('script');
scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>

<style>
    .tableau-container {
        width: 100%;
        overflow: hidden;
        border-radius: 15px;
    }

    .tableauPlaceholder {
        width: 100% !important;
    }

    .tableauViz {
        width: 100% !important;
        min-height: 900px;
    }

    @media (max-width: 768px) {
        .tableauViz {
            min-height: 700px;
        }
    }
</style>

{% endblock %}
```

Dashboard02.html

```
{% extends 'base.html' %}

{% block title %}Dashboard 02{% endblock %}

{% block content %}
```

Heart Disease Analysis using Tableau

```
<div class="container py-5">

  <div class="text-center mb-5">

    <h1 class="fw-bold text-danger">
      HEART DISEASE ANALYSIS Dashboard 02
    </h1>

  </div>

  <div class="tableau-container">
    <div class='tableauPlaceholder' id='viz1781347128632'
      style='position: relative; width:100%; height:900px;'>

      <noscript>
        <a href='#'>
          <img alt='Dashboard02'
src='https://public.tableau.com/static/images/HE/HEARTDISEASEANALYSISUSINGTABLAUE-DASHBOARD02/Dashboard02/1_rss.png'
          style='border:none; width:100%;' />
        </a>
      </noscript>

      <object class='tableauViz' style='display:none;'>
        <param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F' />

        <param name='embed_code_version' value='3' />
        <param name='site_root' value='' />
        <param name='name'
          value='HEARTDISEASEANALYSISUSINGTABLAUE-DASHBOARD02/Dashboard02' />
        <param name='tabs' value='no' />
        <param name='toolbar' value='yes' />
        <param name='static_image'
value='https://public.tableau.com/static/images/HE/HEARTDISEASEANALYSISUSINGTABLAUE-DASHBOARD02/Dashboard02/1.png' />
```

Heart Disease Analysis using Tableau

```
<param name='animate_transition' value='yes' />

<param name='display_static_image' value='yes' />

<param name='display_spinner' value='yes' />

<param name='display_overlay' value='yes' />

<param name='display_count' value='yes' />

<param name='language' value='en-US' />

<param name='filter' value='publish=yes' />

</object>

</div>
</div>

<script type='text/javascript'>

    var divElement = document.getElementById('viz1781347128632');
    var vizElement = divElement.getElementsByTagName('object')[0];

    /* Responsive sizing */
    vizElement.style.width = '100%';

    if (window.innerWidth >= 1200) {
        vizElement.style.height = '900px';
    }
    else if (window.innerWidth >= 768) {
        vizElement.style.height = '800px';
    }
    else {
        vizElement.style.height = '650px';
    }

    var scriptElement = document.createElement('script');
    scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
    vizElement.parentNode.insertBefore(scriptElement, vizElement);

</script>

<style>
    .tableau-container {
        width: 100%;
        overflow: hidden;
        border-radius: 15px;
    }
</style>
```



```
}  
  
.tableauPlaceholder {  
    width: 100% !important;  
}  
  
.tableauViz {  
    width: 100% !important;  
}  
</style>
```

```
        </div>  
    </div>  
</div>  
</div>  
{% endblock %}
```

Story.html

```
{% extends 'base.html' %}  
{% block title %}Story{% endblock %}  
{% block content %}  
  
<div class="container py-5">  
    <div class="text-center mb-5">  
        <div class="tableau-container">  
            <div class='tableauPlaceholder' id='viz1781347729835'  
style='position: relative'>  
                <noscript>  
                    <a href='#'>  
                        <img alt='Story 1'  
  
src='https://public.tableau.com/static/images/HE/HEARTDISEASEANALY  
SISUSINGTABLAUE-STORY/Story1/1_rss.png'  
                        style='border: none; width:100%;' />  
                    </a>  
                </noscript>
```

Heart Disease Analysis using Tableau

```
<object class='tableauViz' style='display:none;'>
  <param name='host_url'
value='https%3A%2F%2Fpublic.tableau.com%2F' />
  <param name='embed_code_version' value='3' />
  <param name='site_root' value='' />
  <param name='name'
value='HEARTDISEASEANALYSISUSINGTABLAUE-STORY/Story1' />
  <param name='tabs' value='no' />
  <param name='toolbar' value='yes' />
  <param name='static_image'
value='https://public.tableau.com/static/images/HE/HEARTDISEASEANA
LYSISUSINGTABLAUE-STORY/Story1/1.png' />
  <param name='animate_transition' value='yes' />
  <param name='display_static_image' value='yes' />
  <param name='display_spinner' value='yes' />
  <param name='display_overlay' value='yes' />
  <param name='display_count' value='yes' />
  <param name='language' value='en-US' />
  <param name='filter' value='publish=yes' />
</object>
</div>
</div>

<script type='text/javascript'>
  var divElement = document.getElementById('viz1781347729835');
  var vizElement = divElement.getElementsByTagName('object')[0];

  // Responsive sizing
  vizElement.style.width = '100%';
  vizElement.style.height = '100vh';

  var scriptElement = document.createElement('script');
  scriptElement.src =
'https://public.tableau.com/javascripts/api/viz_v1.js';
  vizElement.parentNode.insertBefore(scriptElement, vizElement);
</script>

<style>
  .tableau-container {
    width: 100%;
    max-width: 100%;
    margin: 0 auto;
    padding: 0;
  }

  .tableauPlaceholder {
    width: 100%;
```

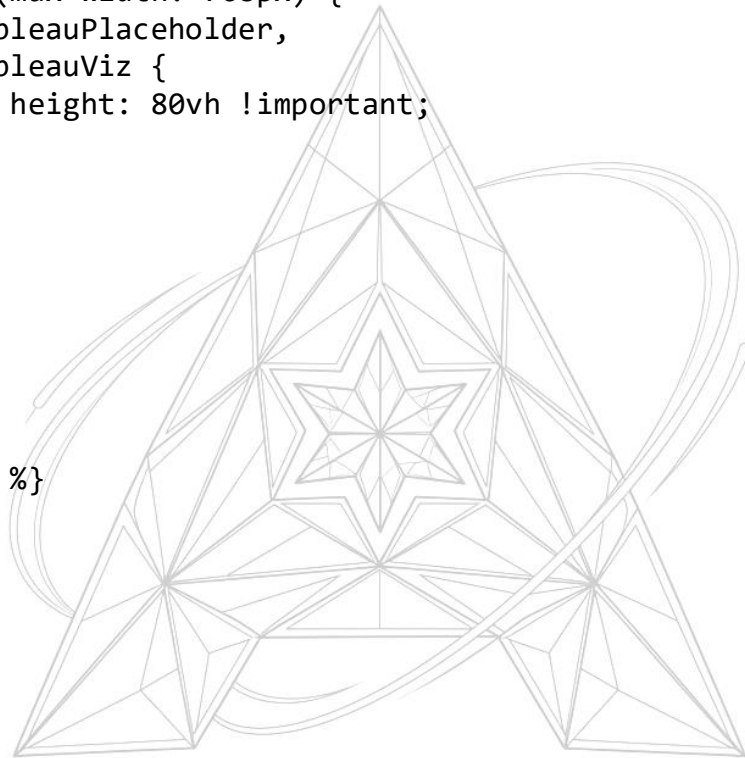
```
        min-height: 100vh;
    }

    .tableauViz {
        width: 100% !important;
        height: 100vh !important;
    }

    /* For smaller devices */
    @media (max-width: 768px) {
        .tableauPlaceholder,
        .tableauViz {
            height: 80vh !important;
        }
    }
</style>

</div>

{% endblock %}
```



ALPHA
— INNOVATORS —
—

